



Spatial Truth: Location Service for Physical AI

APPENDICES FAQ (DEEP DIVES)

APPENDIX C: Financial Projection for 1000-fleet vehicle enterprise

Executive Summary For an enterprise fleet manager operating 1,000 vehicles and generating \$180M-\$200M in annual revenue, “location uncertainty tax” is **\$11M - \$29M** "P&L drag" on their balance sheet.

UC #1: Routing Inefficiency (\$1.2M - \$3.6M Annual Loss): Location uncertainty causes missed turns, circling blocks, and driver overcompensation when map icons "jump."

- The Friction: 10-20-meter drift in dense urban zones triggers 1–3% in total route inefficiency.
- The Impact: At a fleet revenue baseline of \$180M, this represents a annual loss of \$1.8M - \$5.4M.
- The Recovery: Attributing a ~70% slice to positioning errors, centimeter accuracy recovers \$1.2M-\$3.6M annually in direct routing waste.

UC #2: Yard & Warehouse Dwell (\$800K - \$1.5M Annual Loss): Poor location prevents automated arrivals and dock assignments, inflating "yard dwell" by 8-15 minutes per stop.

- The Friction: 3 stops/day × 10 minutes lost = 500 fleet-hours lost per day.
- The Cost: At \$50-\$75 per vehicle-hour,
- The Daily waste: 500 fleet-hours X \$50 (low end) = \$25,000 per day on wasted time.
- Annual Waste: \$25,000/day X 250 workdays = ~ \$6M/year. (250 workdays per year excluding weekends/holidays).
- The Recovery: By enabling accuracy automated check-ins, Spatial Truth eliminates the 10–15% of dwell time caused by location uncertainty, saving \$800K-\$1.5M per year.

UC #3: Asset Utilization Drag (\$9M - \$21.6M Annual Loss): When telemetry is unreliable, dispatchers cannot reassign jobs with confidence, leaving assets productive for fewer hours per day.

- The Friction: Multi-city fleets report a 5-12% utilization drag due to low location accuracy.
- The Impact: With each truck producing \$180,000/year, a 5% loss equals \$9,000 per truck.
- The Scale: Across 1,000 vehicles, this represents \$9M-\$21.6M in unrealized revenue.

UC #4: Insurance & Liability Exposure (\$650K - \$2M Avoidable Expense): Insurers penalize fleets with "lost telemetry" segments or ambiguous liability evidence caused by GPS jitter during collisions.

- The Friction: Average insurance costs range from \$6,500-\$10,000 per vehicle.
- The Impact: Location uncertainty during safety events contributes to a 10-20% premium increase.
- The Recovery: Providing a sub-meter "source of truth" for collision logs can reduce annual fleet premiums by \$650,000-\$2,000,000.

UC #5: Autonomy Stalling (Frozen CAPEX & Delayed ROI): Level 2.5 and Level 3 systems require sub-meter localization for lane-level guidance and safety compliance.

- The Friction: When GPS drifts 10-20-meters, autonomous systems disengage, forcing a fallback to expensive "Hardware Clutter" (LiDAR/RTK).
- The Impact: This delays the ROI of multi-million-dollar autonomy programs and keeps capital idle.
- The Solution: Network-level centimeter accuracy allows these programs to scale beyond test loops into revenue-generating urban routes.

*Table 2.0: Location Error Total Annual Loss for 1,000-Vehicle Fleets **\$11M - \$29M***

Use Case	Low-End Loss	High-End Loss
UC #1: Routing Inefficiency	\$1.2 Million	\$3.6 Million
UC #2: Yard & Warehouse Dwell	\$0.8 Million	\$1.5 Million

Use Case	Low-End Loss	High-End Loss
UC #3: Asset Utilization Drag	\$9.0 Million	\$21.6 Million
UC #4: Insurance & Liability	\$0.65 Million	\$2.0 Million
TOTAL ANNUAL LOSS	\$11.65 Million	\$28.7 Million